

# Hybrid\_v3

## Phase 4 — Dynamic Model Selection

### Complete Technical Progress Report

#### V3.9 → V3.12 → V3.13

## Nested OOF Selection, Conditional Selection & Behavioral Diagnostics

**Dataset:** ISIC 2019

**Evaluation set:** 1,267 leakage-safe test samples

**Candidate models:** EfficientNet-v2-S, ResNet50, MobileNet-v3-Large

**Objective:** learn when an alternative model can correct an EfficientNet error without sacrificing its strong baseline.

Prepared from the Phase 4 experiments completed through V3.13.

# 1. Executive Summary

Phase 4 investigates dynamic model selection rather than relying on a single classifier or static fusion. EfficientNet-v2-S is the strongest individual baseline, but ResNet50 and MobileNet-v3-Large contain complementary correct predictions. The research question became whether these conditional rescue opportunities can be detected from inference-time model signals.

The work progressed through nested pairwise selection (V3.9), conditional-selection diagnostics (V3.11), a leakage-controlled conditional selector (V3.12), and selector behavioral analysis (V3.13).

| Stage | Purpose                                         | Result                                                |
|-------|-------------------------------------------------|-------------------------------------------------------|
| V3.9  | Nested pairwise OOF foundation                  | Generated fold-safe OOF prediction/decision signals   |
| V3.11 | Test whether conditional selection is learnable | ResNet rescue AUC 0.9464; MobileNet rescue AUC 0.9582 |
| V3.12 | Train conditional selector                      | 74.4278% → 77.1902%                                   |
| V3.13 | Analyze behavior and failures                   | 73 rescues, 38 harms, 29 neutral transitions          |

**Headline:** V3.12 improves the EfficientNet baseline by **+2.7624 percentage points** under nested outer OOF evaluation. Oracle accuracy is 87.4507%.

## 2. Experimental Context

The Phase 4 evaluation uses 1,267 ISIC 2019 test samples. The principal dynamic-selection candidates are the best baseline model plus two complementary CNNs.

| Model              | Accuracy | Correct / 1267 |
|--------------------|----------|----------------|
| EfficientNet-v2-S  | 74.4278% | 943            |
| ResNet50           | 73.8753% | 936            |
| MobileNet-v3-Large | 73.6385% | 933            |

The important observation is not the small difference in standalone accuracy, but the complementary error structure: an alternative can sometimes be correct when EfficientNet is wrong.

## 3. V3.9 — Nested Pairwise Selection Foundation

V3.9 established the nested OOF foundation used by later experiments. Its 1,267-row output contains predictions, correctness labels for evaluation/target construction, confidence scores, gate probabilities, rescue/harm probabilities, net scores, decisions and policy thresholds.

The key methodological property is nested evaluation: decisions are generated without optimizing the policy on the held-out outer-fold labels. This output became the canonical source for the V3.11 diagnostic and V3.12 selector.

## 4. V3.11 — Conditional Selection Diagnostic

V3.11 was diagnostic-only. It trained no selector and optimized no final threshold. Its purpose was to determine whether conditional rescue and harm events could be detected from current OOF features.

### 4.1 Correctness patterns

| Pattern  | Count | Share  |
|----------|-------|--------|
| E0_R0_M0 | 159   | 12.55% |
| E0_R0_M1 | 46    | 3.63%  |
| E0_R1_M0 | 69    | 5.45%  |
| E0_R1_M1 | 50    | 3.95%  |
| E1_R0_M0 | 45    | 3.55%  |
| E1_R0_M1 | 81    | 6.39%  |
| E1_R1_M0 | 61    | 4.81%  |
| E1_R1_M1 | 756   | 59.67% |

E/R/M denote EfficientNet, ResNet and MobileNet correctness. The key population is E0: EfficientNet errors.

## 4.2 Rescue/harm populations

| Event            | Count |
|------------------|-------|
| ResNet rescue    | 119   |
| ResNet harm      | 126   |
| MobileNet rescue | 96    |
| MobileNet harm   | 106   |

## 4.3 Rescue detectability

| Target        | Feature                       | AUC    |
|---------------|-------------------------------|--------|
| ResNet rescue | resnet_gate                   | 0.9464 |
| ResNet rescue | resnet_conf_gate_minus_effnet | 0.9462 |
| ResNet rescue | resnet_conf_x_gate            | 0.9458 |
| Mobile rescue | mobile_gate                   | 0.9582 |
| Mobile rescue | mobile_conf_x_gate            | 0.9568 |
| Mobile rescue | mobile_conf_gate_minus_effnet | 0.9509 |
| ResNet harm   | resnet_agrees_effnet          | 0.9238 |
| Mobile harm   | mobile_agrees_effnet          | 0.9324 |

These high AUCs were the major justification for moving from diagnosis to learned conditional selection. Rescue and harm events are structured rather than random.

## 4.4 Theoretical headroom

| Metric                | Value       |
|-----------------------|-------------|
| EfficientNet baseline | 74.4278%    |
| Oracle                | 87.4507%    |
| Theoretical headroom  | +13.0229 pp |
| EfficientNet errors   | 324         |
| Recoverable errors    | 165         |
| Unrecoverable errors  | 159         |

## 5. V3.12 — Conditional Model Selector

V3.12 converts the V3.11 evidence into a learned selector among EfficientNet-v2-S, ResNet50 and MobileNet-v3-Large. Ground-truth correctness is used only to construct targets and evaluate held-out outer folds; correctness-derived information is not supplied as a selector feature.

### 5.1 Target distribution

| Target             | Count | Share  |
|--------------------|-------|--------|
| EfficientNet-v2-S  | 1102  | 86.98% |
| ResNet50           | 119   | 9.39%  |
| MobileNet-v3-Large | 46    | 3.63%  |

### 5.2 Feature matrix

The final matrix contains 33 selector features. The most important features are model agreement, prediction diversity, gate probabilities, confidence/gate interactions and relative gate differences.

| Rank | Feature                | Importance |
|------|------------------------|------------|
| 1    | all_agree              | 0.113810   |
| 2    | resnet_agrees_mobile   | 0.106780   |
| 3    | num_unique_predictions | 0.097675   |
| 4    | resnet_agrees_effnet   | 0.094901   |
| 5    | mobile_agrees_effnet   | 0.075405   |
| 6    | max_gate               | 0.065185   |
| 7    | resnet_gate            | 0.051353   |
| 8    | resnet_conf_x_gate     | 0.045957   |
| 9    | gate_difference        | 0.042320   |
| 10   | resnet_gate_minus_conf | 0.033285   |

### 5.3 Outer nested results

| Fold | Base     | V3.12    | Gain       | Transitions | Rescues | Harms |
|------|----------|----------|------------|-------------|---------|-------|
| 1    | 72.0472% | 74.0157% | +1.9685 pp | 32          | 17      | 12    |
| 2    | 73.2283% | 78.7402% | +5.5118 pp | 27          | 17      | 3     |
| 3    | 75.8893% | 77.8656% | +1.9763 pp | 29          | 13      | 8     |
| 4    | 76.2846% | 79.0514% | +2.7668 pp | 17          | 10      | 3     |
| 5    | 74.7036% | 76.2846% | +1.5810 pp | 35          | 16      | 12    |

### 5.4 Final V3.12 result

| Metric              | Value      |
|---------------------|------------|
| Baseline            | 74.4278%   |
| V3.12 selector      | 77.1902%   |
| Absolute gain       | +2.7624 pp |
| Oracle              | 87.4507%   |
| Transitions         | 140        |
| Rescues             | 73         |
| Harms               | 38         |
| Neutral transitions | 29         |
| Rescue/harm ratio   | 1.9211     |

V3.12 therefore demonstrates a meaningful improvement over the best single model while preserving a fully nested evaluation design.

## 5.5 Selected model distribution

| Model              | Count | Share  |
|--------------------|-------|--------|
| EfficientNet-v2-S  | 1127  | 88.95% |
| ResNet50           | 108   | 8.52%  |
| MobileNet-v3-Large | 32    | 2.53%  |

## 6. V3.13 — Selector Behavior Analysis

V3.13 analyzes where V3.12 succeeds and fails. It is not another training stage; it is the diagnostic basis for the next selector design.

### 6.1 Transition breakdown

| Transition               | Count | Rescues | Harms | Neutral | Net gain | Accuracy |
|--------------------------|-------|---------|-------|---------|----------|----------|
| EfficientNet → ResNet    | 108   | 56      | 32    | 20      | +24      | 51.85%   |
| EfficientNet → MobileNet | 32    | 17      | 6     | 9       | +11      | 53.13%   |
| Total                    | 140   | 73      | 38    | 29      | +35      | —        |

### 6.2 Rescue characteristics

| Transition         | Rescues | Selection conf. | EffNet conf. | Alternative gate |
|--------------------|---------|-----------------|--------------|------------------|
| EffNet → ResNet    | 56      | 0.8998          | 0.7564       | 0.8992           |
| EffNet → MobileNet | 17      | 0.8230          | 0.8199       | 0.8998           |

### 6.3 Harm characteristics

| Transition         | Harms | Selection conf. | EffNet conf. | Diagnostic signal           |
|--------------------|-------|-----------------|--------------|-----------------------------|
| EffNet → ResNet    | 32    | 0.8170          | 0.8063       | ResNet gate high            |
| EffNet → MobileNet | 6     | 0.6883          | 0.9264       | Alternative margin negative |

A key finding is that selector confidence is not a safety guarantee. Harmful EfficientNet→ResNet switches still had mean selection confidence 0.8170.

### 6.4 Class-wise performance

| Class | N   | Base    | V3.12   | Δ        |
|-------|-----|---------|---------|----------|
| DF    | 11  | 27.27%  | 36.36%  | +9.09 pp |
| BKL   | 132 | 64.39%  | 68.94%  | +4.55 pp |
| BCC   | 167 | 75.45%  | 79.04%  | +3.59 pp |
| SCC   | 31  | 29.03%  | 32.26%  | +3.23 pp |
| NV    | 644 | 85.09%  | 87.89%  | +2.80 pp |
| MEL   | 226 | 63.27%  | 65.04%  | +1.77 pp |
| VASC  | 12  | 100.00% | 100.00% | 0.00 pp  |
| AK    | 44  | 38.64%  | 36.36%  | -2.27 pp |

AK is the clearest class-level regression and should receive focused analysis. Small classes such as DF, SCC and VASC require cautious interpretation because of low sample counts.

### 6.5 EfficientNet error recovery

| Category           | Count | Fraction of EffNet errors |
|--------------------|-------|---------------------------|
| ResNet-only rescue | 69    | 21.30%                    |
| Mobile-only rescue | 46    | 14.20%                    |
| Both rescue        | 50    | 15.43%                    |
| Neither rescue     | 159   | 49.07%                    |
| Recoverable total  | 165   | 50.93%                    |

V3.12 captures only 21.21% of the theoretical recoverable headroom. This is the principal remaining opportunity.

## 7. What We Learned

**Conditional selection is learnable.** V3.11's high rescue AUCs translated into a real nested accuracy gain in V3.12.

**Gate probabilities matter but are insufficient.** They are among the strongest signals, yet high gate values can still lead to harm.

**Agreement is a major signal.** all\_agree, pairwise agreement and prediction diversity dominate feature importance, showing that model disagreement contains useful information.

**Confidence alone is insufficient.** Rescue confidence is higher on average than harm confidence, but the distributions overlap.

**Large headroom remains.** EfficientNet has 324 errors; 165 are theoretically recoverable by the three-model pool, while 159 are errors for all three.

## 8. Leakage / Evaluation Audit

| Property                                 | Status                      |
|------------------------------------------|-----------------------------|
| Nested outer evaluation                  | Yes                         |
| Inner selector/threshold selection       | Inside outer training folds |
| Correctness-derived selector feature     | No                          |
| Correctness used for target construction | Yes                         |
| Final nested OOF samples                 | 1,267                       |

The V3.12 result should therefore be treated as the principal Phase 4 experimental result, subject to the usual limitations of a single fixed evaluation set.

## 9. Phase 4 Output Artifacts

**V3.11:** conditional features, feature separation, rescue AUC analysis, confidence bins, class-wise analysis, disagreement analysis, headroom JSON, conditional case populations and summary.

| V3.12 file                         | Purpose                      |
|------------------------------------|------------------------------|
| v3_12_feature_matrix.csv           | 1,267 × 33 selector features |
| v3_12_feature_importance.csv       | Selector feature importance  |
| v3_12_fold_metrics.csv             | Five outer-fold metrics      |
| v3_12_model_comparison.csv         | Baseline vs selector         |
| v3_12_selected_models.csv          | Selected model records       |
| v3_12_selector_oof_predictions.csv | Final nested OOF predictions |
| v3_12_summary.json                 | Machine-readable summary     |

**V3.13:** analysis table, transition analysis, rescue/harm breakdowns, class-wise performance, confidence/gate/selection bins, EfficientNet error analysis, recoverable headroom, fold stability, ranked rescue/harm cases and summary.

## 10. Recommended Next Experiment — V3.14

- 1. Explicit switch-vs-keep policy.** Instead of only predicting the globally preferred model, directly estimate whether leaving EfficientNet is beneficial.
- 2. Separate ResNet and MobileNet policies.** ResNet accounts for 56 of 73 observed rescues, while MobileNet accounts for 17; their conditional behavior is different.
- 3. Add an abstain/keep-EfficientNet option.** The selector should be able to refuse a switch when evidence is weak.
- 4. Calibrate selector confidence.** High-confidence harms show that raw selection probability is not yet a sufficient safety criterion.
- 5. Add class-aware analysis/features.** The AK regression and class-dependent gains justify class-conditional policies without using ground-truth labels as inputs.
- 6. Optimize rescue/harm utility.** Track accuracy, rescue rate, harm rate, rescue/harm ratio, transition rate and captured recoverable headroom together.

## 11. Final Phase 4 Conclusion

Phase 4 has established the full experimental chain: **model complementarity** → **conditional diagnostics** → **nested selector training** → **behavioral failure analysis**.

The strongest result is V3.12: **74.4278% → 77.1902%**, a **+2.7624-point** improvement, with 73 rescues and 38 harms across 140 transitions.

V3.13 shows that the result is not yet optimal: only **21.21%** of the theoretical recoverable headroom is captured. The next challenge is therefore not proving complementarity, but making the switch decision more precise and safer.

**Scientific conclusion:** model complementarity is real and conditionally detectable, but the current selector is conservative and imperfect at distinguishing beneficial switches from harmful switches.

**Current best:** V3.12 nested conditional selector — 77.1902%.

**Oracle:** 87.4507%.

**Next target:** increase recoverable-headroom capture while reducing harmful transitions.